

Term Project: Personal Data Visualization

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Course: Programing Foundations

1. Data Collection Rationale and Process

The goal of this project was to track my cooking habits to understand the relationship between my emotional state (mood), the taste of the dishes (mastery), and the time spent in the kitchen.

I recorded data manually after every meal preparation using a spreadsheet.

Parameters: I tracked four main variables:

- **Dish Name:** Type of meal prepared.
- **Time:** Duration in minutes.
- **Mastery Level:** A self-assessed scale from 1 to 5 based on my comfort with the recipe.
- **Mood:** A scale from 1 (Stressed/Low energy) to 5 (Inspired/High energy).
- **Timeline:** Data was collected consistently from early January 2026 until March 25, 2026.

2. Final Visualization Rationale

I chose a scatterplot inspired layout where the X-axis represents my Mood and the Y-axis represents Time.

- **Shapes (Mastery):** To represent the level of mastery, I designed custom SVG polylines. For example, a simple rectangle represents Level 1, while more complex geometric shapes (like a "T" or a "Z") represent higher levels of expertise. This allows the viewer to know the quality of the food.
- **Color (Categories):** I used a categorical color scale to differentiate types of food (Meat, Legumes, Pasta, Soup). This helps identify if certain food groups are prepared faster than others.

- **Tools:** The visualization was built using **D3.js (v5.12)**, utilizing linear scales to map data values to pixel coordinates accurately.

3. Alternate Idea & Comparison

Before finalizing the design, I explored three distinct conceptual approaches (Sketch 1, 2, 3 attached at the end of the document) to determine which would most effectively communicate the relationship between my emotional state and my cooking habits.

- **Sketch 1: Radial/Circular Chart**

This sketch explored a semi-circular layout where data points were arranged in concentric layers. It was an experimental way to visualize the passage of time or categories.

It was visually unique; it made it very difficult to read the exact correlation between two numerical variables (Mood and Time). The radial geometry distorted the distance between points, making the data less accessible for a quick analysis.

- **Sketch 2: Coordinates**

A clean, traditional coordinate system (X-Y) using standard markers to plot the data.

It was highly functional and easy to read. However, it felt too generic for a personal data project. It lacked a way to represent the Mastery level effectively without adding a third axis or making the chart overly complex with text labels.

- **Sketch 3: Tetris design**

I chose to implement **Sketch 3** because it combined the efficiency of the Cartesian coordinate system from N2 with a creative way to encode the Mastery variable.

Inspired by the geometric blocks in Sketch 3, I developed custom polylines for each mastery level. A simple shape represents Level 1, while more complex, "Tetris" shapes (L-shapes, T-shapes, etc.) represent higher levels of mastery.

Visual Encoding: This allows the viewer to see three dimensions of data at once: Position (Time vs. Mood), Color (Food Category), and Shape Complexity (Mastery).

Conclusion: This design was the most successful because it turned abstract numbers into recognizable geometric shapes, making the effort spent on complex recipes visually apparent.

4. Conclusions Based on Data

After analyzing the final visualization, I discovered a clear trend: when my **Mood** is high (4 or 5), I tend to spend more time cooking complex dishes (Mastery 4-5). Conversely, on days when my mood was low, I gravitated toward "Level 1" meals or eating out (Time = 0). This suggests that for me, cooking is an activity fueled by creative energy rather than a purely functional chore.

5. Conclusions Based on Data

Reflecting on the past twelve weeks of data collection, the process has provided significant insights into my personal habits, but it also revealed the inherent challenges of manual self-tracking.

Challenges and Lessons Learned: One of the primary difficulties was maintaining consistency during busy academic weeks. I noticed that on days with high stress, I was more likely to forget to record my "Mood" or "Time" immediately after cooking, which required me to rely on memory later that evening. This taught me that the accuracy of personal data is highly dependent on the friction of the collection method. Additionally, categorizing "Mood" on a simple 1-5 scale sometimes felt reductive, as it couldn't capture the nuance between "tired but happy" versus "energetic but stressed."

Future Improvements: If I were to extend this project, I would implement the following changes to improve data integrity and depth:

- **Passive Data Collection:** To eliminate memory discrepancies, I would use a dedicated kitchen timer or a smart-home integration to automatically log the start and end times of my cooking sessions.
- **Environmental Context:** I would add a variable for "External Factors," such as university deadlines or weather conditions. I suspect that cold weather in Vancouver directly correlates with my "Soup" category and longer preparation times.

- **Nutritional Correlation:** A future iteration could link this dataset with nutritional value to see if higher "Mastery" and "Time" investment results in healthier eating habits compared to quick "Level 1" meals.

Ultimately, this project has transformed how I view my time in the kitchen, not just as a chore, but as a measurable reflection of my wellbeing and creative energy.

6. Dataset

Date	Dish	Time(min)	Mastery(1-5)	Mood(1-5)
25/01	Chicken fajitas	41 min	3.5 / 5	4
26/01	Lima Beans / burger / rice	90:32 min	3 / 5	2
27/01	Lentils / pork / rice	70:16 min	4 / 5	5
28/01	Lentils / pork (Leftovers)	10 min	4 / 5	1
29/01	Quinoa and chicken	34 min	5 / 5	5
30/01	Chicken Salad	23 min	3 / 5	4
31/01	Spaghetti Bolognese	20 min	5 / 5	3
01/02	Soup (Out)	N/A	N/A	-
02/02	Burgers / rice / fries	25 min	4 / 5	2
03/02	Tuna with rice	23 min	3 / 5	2
04/02	Soup (Home)	38 min	3 / 5	3
05/02	McDonalds	N/A	N/A	-
06/02	Spaghetti Bolognese	20 min	5 / 5	4
07/02	Tuna fajitas	20 min	5 / 5	4
08/02	Tuna salad	15 min	5 / 5	2
09/02	Lentils / pork / rice	60 min	3 / 5	2
10/02	Chicken / Rice / Salad	18 min	5 / 5	4
11/02	Quinoa and chicken	38 min	5 / 5	2
12/02	Spaghetti Bolognese	26min	5 / 5	4
13/02	Lima Beans / burger / rice	60 min	5 / 5	4
14/02	Chicken Nuggets Fajitas	23 min	4 / 5	4
15/02	-	-	-	-
16/02	McDonalds	N/A	N/A	-
17/02	Chicken / Rice / Salad	22 min	3 / 5	2
18/02	Red Beans/ Rice / Pork	35 min	4 / 5	4
19/02	Spaghetti Bolognese	18 min	4 / 5	2
20/02	Burgers / rice / fries	30 min	5 / 5	3
21/02	Wendy's	N/A	N/A	-
22/02	Quinoa and chicken	27 min	5 / 5	4
23/02	-	-	-	-
24/02	-	-	-	-
25/02	Tuna with rice	15 min	4 / 5	2
26/02	Red Beans/ Rice / Pork	35 min	5 / 5	4
27/02	A&W	N/A	N/A	-
28/02	Chicken fajitas	27 min	3.5 / 5	4
01/03	Tuna fajitas	15 min	4 / 5	4
02/03	Burger / rice	30 min	4 / 5	3
03/03	chickpeas / pork / rice	33 min	3 / 5	4
04/03	Spaghetti Bolognese	20 min	3 / 5	2
05/03	Burger & Salad	28 min	3 / 5	4
06/03	Chicken / Rice / Salad	20 min	5 / 5	4
07/03	Tuna fajitas	24 min	4 / 5	2
08/03	-	-	-	-

7. References

D3.js Documentation: <https://github.com/d3/d3/blob/v5.12.0/API.md>

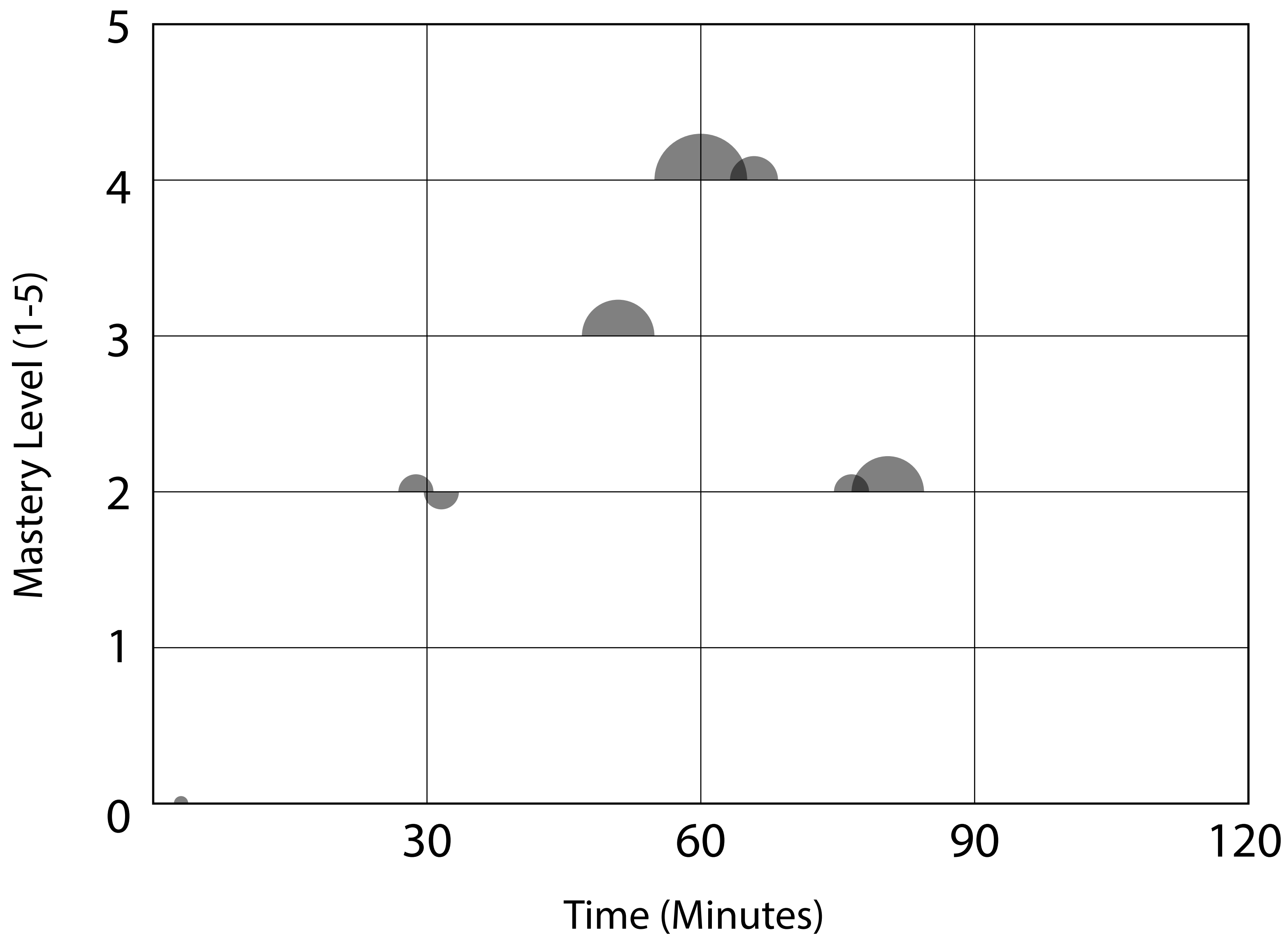
Data to Viz (Scatter plot): <https://www.data-to-viz.com/graph/scatter.html>

Class Lectures & Lab Materials: Concepts for D3.json data loading and promise handling were adapted from course lab exercises.

8. Project Directory

- Final Project: <https://taynus.github.io/dgm6109/term/final>
- Draft 1: <https://taynus.github.io/dgm6109/term/draft1/>
- Draft 2: <https://taynus.github.io/dgm6109/term/draft2/>

Sketch 1

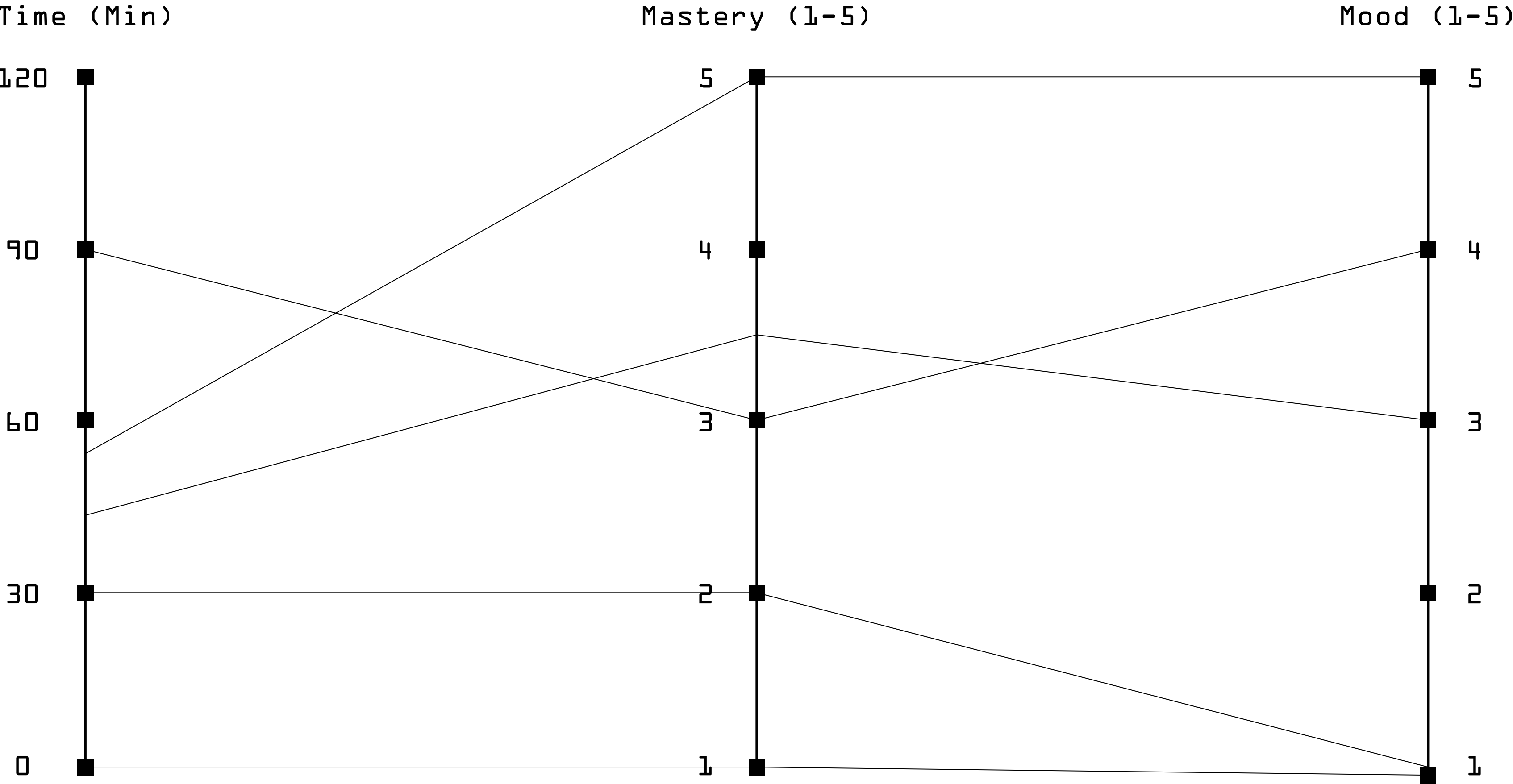


Mood Score

- | | | |
|---|---|---|
|  | 1 | I do a lot the last day and I just reheated |
|  | 2 | Not too inspired |
|  | 3 | Try to discover new flavors |
|  | 4 | Creative to try by myself |
|  | 5 | Inspired to cook some complex dish |

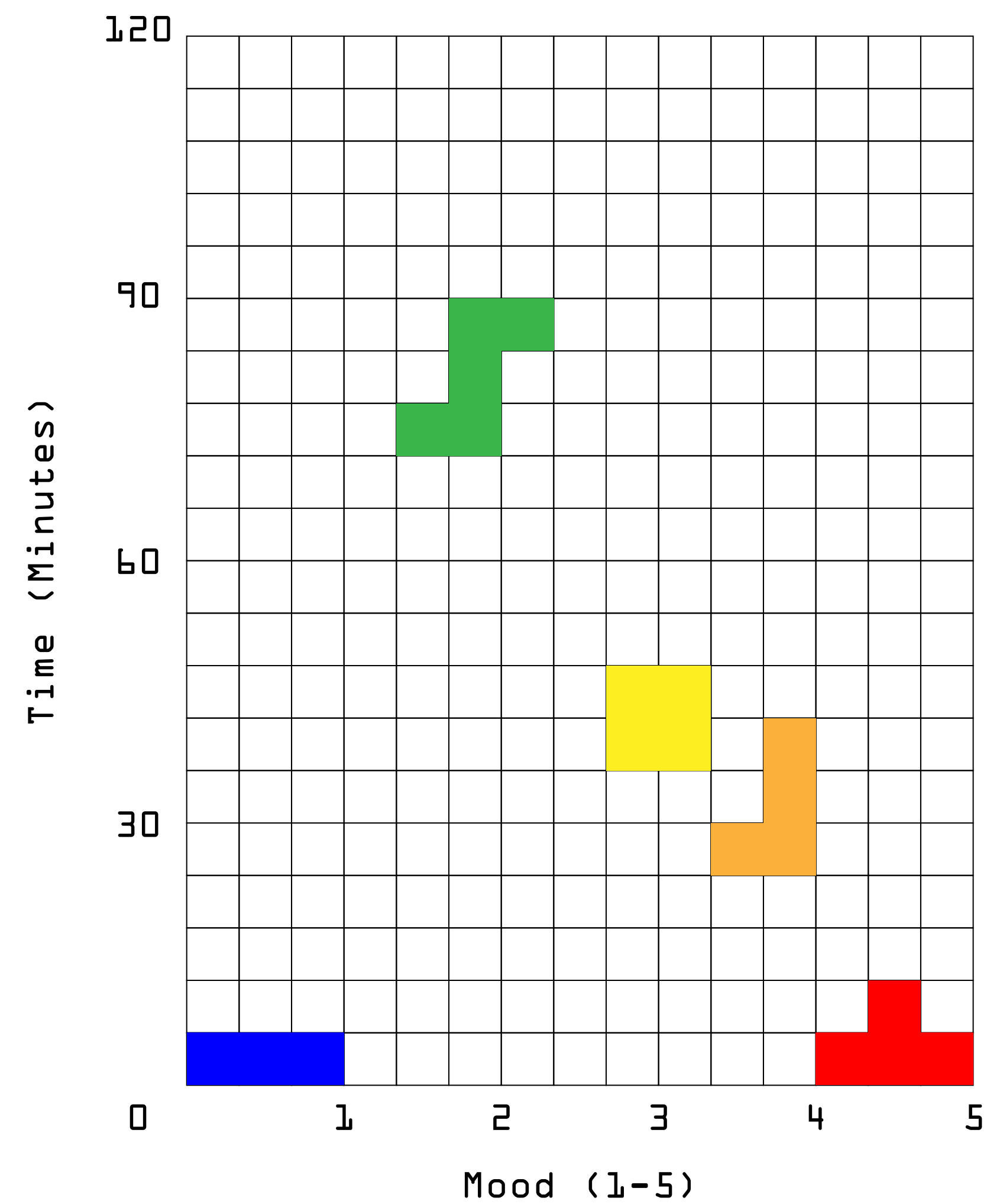
Large circles show better mood

Sketch 2



Legend	
	Individual cooking session line

Sketch 3



Mastery
Level (1-5)

